Varchini Devi K

Senior Software Engineer

Chennai, India | +91 80564 37139 | varshinikannan2495@gmail.com
LinkedIn | GitHub | Portfolio

SUMMARY

Senior Software Engineer with 8+ years building and sustaining high-performance, security-sensitive enterprise platforms in C#, ASP.NET Core / .NET 8, Web API and Azure, across 11 enterprise applications. Currently on a digital forensics and eDiscovery platform, working on evidence collection, endpoint and mobile integration, performance at scale and a redesign of evidence processing. Experienced in modernising legacy .NET Framework systems, event-driven Azure workflows and Angular / React front ends. Over the past year I have also built ten AI systems (RAG, hybrid retrieval, tool-calling and multi-agent systems, MCP, inference routing) with evaluation, guardrails and cost tracking.

CORE SKILLS

Backend: C#, ASP.NET Core / .NET 8, Web API, ADO.NET, Entity Framework Core, LINQ, WCF, ASP.NET MVC / Web Forms, VB.NET, Python

Cloud and DevOps: Azure (Functions, Durable Functions, Logic Apps, Service Bus, Key Vault, Managed Identity, IAM, Blob Storage, Cosmos DB, App Configuration, Application Insights), AWS EC2, Docker, AKS, Azure DevOps, Jenkins, CI/CD

Front end: Angular, React, Next.js, TypeScript, JavaScript, HTML, CSS, Bootstrap

Data: SQL Server, PostgreSQL / pgvector, MongoDB, DB2, Cosmos DB

Security: OAuth 2.0, OpenID Connect, JWT, certificate-based encryption, Managed Identities

Applied AI: RAG, hybrid search, tool-calling agents, LangGraph, MCP, evaluation (LLM-as-judge), guardrails, GitHub Copilot, Claude Code

Mobile (personal projects): Kotlin / Jetpack Compose, React Native / Expo

EXPERIENCE

Exterro R&D Private Ltd - Senior Software Engineer

Aug 2024 - Present

FTK Central: eDiscovery and digital forensics platform used by legal teams, corporate investigators and law enforcement. Worked across most of the application: case creation, collection, evidence processing, review and live preview.

- Lead the migration from .NET Framework to .NET 8: resolving breaking changes, refactoring legacy components and modernising APIs for long-term support.
- Helped lead the redesign of evidence processing to a database-driven, horizontally scalable job model, with centralised job management, synchronised status and a case-level job screen showing alerts, errors and adjustable job priority.
- Implemented evidence collection from computers, mobile devices, Gmail, Outlook, Exchange, Microsoft Teams, Slack, Zoom, Google Drive, OneDrive and network shares, each through its own route: endpoint agents and a site server, or the source's own APIs (Microsoft Graph, Google APIs, Export APIs).
- Implemented case creation, evidence processing and review workflows and an on-demand live preview that downloads files only when they are opened; engineered the mobile collection integration end to end (job orchestration, secure transfer, progress tracking, encrypted output).
- Designed secure connectivity and file transfer between distributed agents and central servers across domains, networks and regions, using VPN tunnelling, dynamic address handling and HTTPS-based transfer.
- Developed and sustained core platform modules in C#, ASP.NET / .NET, Web API, Angular 14, EF Core, ADO.NET, LINQ and SQL Server in an Agile, globally distributed team; implemented certificate-based encryption for forensic data integrity; integrated Apple APIs and macOS / iWork handling; enabled automated entity extraction and classification; designed Angular UI for evidence review, filtering, tagging and entity linking.
- Reduced the load time of a high-volume job management screen from over 10 minutes to 3-6 seconds through query, code and database optimisation.
- Resolved critical production issues, including hangs under high collection concurrency, timeouts and thread contention when a remote collection component was unavailable, and timeouts on mail servers under heavy load; debugged complex scenarios across services, data pipelines and integrations; handled corrupted data from diverse sources.
- Resolved client-reported collection issues across cloud and on-premises sources; optimised SQL stored procedures and data models, improving correlation accuracy and reducing processing time by about 35%.

Technologies: C#, .NET 8, ASP.NET, Web API, Angular 14, EF Core, ADO.NET, LINQ, SQL Server

Wipro Limited - Senior Software Engineer

Feb 2022 - Jul 2024

Two client engagements: Halliburton (digital oilfield solutions on Azure) and HP Marcoms (enterprise web applications and platform migration).

Halliburton

- Developed and enhanced digital oilfield solutions for drilling data processing, asset tracking and job scheduling, improving operational efficiency; delivered mobile-support APIs for well data access.
- Built RESTful APIs and microservices with .NET Core, C# and Entity Framework; designed Angular 14 dashboards for real-time telemetry and operations; integrated React / Next.js front ends with versioned .NET REST APIs.
- Designed event-driven workflows with Azure Blob Storage and Durable Orchestrations: file drops triggered automated decryption, hash-based duplicate detection, validation and splitting, then structured SQL Server insertion; Azure Functions with action triggers orchestrated sequential inserts for consistency and reliability.

- Used system- and user-assigned Managed Identities across Function Apps, Blob Storage, Cosmos DB and Key Vault, replacing stored secrets and connection strings; secured certificates in Key Vault; worked with Azure IAM and Logic Apps.
- Built Cosmos DB container models for UI retrieval (partitioning, indexing, RU tuning); implemented Service Bus topics and subscriptions with retry policies and dead-letter queues; centralised configuration and feature toggles in App Configuration.
- Monitored with Application Insights (custom telemetry, dependency tracking, KQL); built and supported Dockerised services on Azure Kubernetes Service; used Storage Explorer and the Storage Emulator for local testing.

HP Marcoms

- Led end-to-end development of 3+ applications (requirements, design, coding, testing, deployment, production support) independently; mentored team members on Agile compliance and timely sprint delivery.
- Built dynamic web apps with ASP.NET MVC / Core and React, improving responsiveness and load times; designed and optimised complex SQL queries, stored procedures and UDTs; developed secure REST APIs with SSL and HTTPS compliance.
- Upgraded applications to .NET 4.8 and migrated Windows Server 2012 to 2019 with full environment setup and validation; migrated applications to AWS EC2, reducing cost by 40% and improving scalability.
- Tuned IIS for high-traffic applications, improving response time by 30%; managed 100+ scheduled jobs in Tidal, reducing system downtime by 40%.
- Automated CI/CD with Jenkins and Azure DevOps; ran Agile workflows in Azure Boards; delivered client-specific features and fixes, coordinated UAT and secured production sign-offs; partnered with cybersecurity to maintain a 100% clean security record.

Technologies: .NET Core, ASP.NET MVC, C#, Angular 14, React / Next.js, Azure Functions, Logic Apps, Service Bus, Key Vault, Cosmos DB, SQL Server, Docker, AKS, Jenkins, Azure DevOps

Cognizant Technology Solutions - Associate Feb 2018 - Feb 2022

Insurance applications for Travelers: claims, subrogation and payments.

- Led the full SDLC (requirements, design, coding, testing, deployment, support, maintenance, enhancements, documentation) using Visual Studio .NET, MongoDB, DB2 and SQL Server per client standards.
- Designed and built new UI screens from scratch with ASP.NET, C#, VB.NET, ADO.NET, Web API, JavaScript, SQL Server, DB2 and MongoDB; built the Adverse Subrogation screen, improving efficiency in managing complex legal cases; added AJAX-based dynamic content loading and validation.
- Used MongoDB for dynamic case data behind high-performance Web APIs that reduced case access and update time; secured APIs with OAuth and JWT; tested with Postman and Swagger.
- Integrated a secure online payment system for Travelers Insurance to streamline claim payments and refunds.
- Implemented LINQ queries, WCF services and complex ASP.NET Web Forms using C# and .NET Framework 4.0; used ADO.NET for high-volume data retrieval, storage and manipulation.
- Managed version control with GitHub and SVN; automated CI/CD with Jenkins and integrated Urban Code Deploy for end-to-end build, test and deployment automation, increasing release frequency.
- Designed and ran test strategies across environments; conducted live demos for business users; took part in planning, design and code reviews, security requirements, data-flow validation, status reporting and all Agile ceremonies.

Technologies: ASP.NET, C#, VB.NET, Web API, WCF, ADO.NET, LINQ, MongoDB, DB2, SQL Server, Jenkins, Urban Code Deploy

RECOGNITION

- 15+ internal excellence awards, including Best Employee and the Bright Beginning Award, for high-impact delivery, ownership and client satisfaction.
- Drove modernisation initiatives by learning and adopting new frameworks and cloud services beyond assigned responsibilities.
- Trusted by managers and clients for reliability, collaboration and on-time delivery across complex engagements.

SELECTED INDEPENDENT PROJECTS (2026)

- **AdviShopping** - Hybrid search, visual search, a tool-calling shopping agent and validated document extraction. Next.js, Supabase / pgvector. Live demo
- **inferouter** - Cost-aware LLM inference router with a confidence cascade across self-hosted and larger models. Python, FastAPI. Source
- **AI News Agent and AyvaBot** - Scheduled LLM news filter on GitHub Actions; personal multi-agent assistant with approval gates and fact-checking. Source
- **GitHub Agent** - MCP server (22 tools), LangGraph multi-agent team, hybrid RAG, approval-gated writes, CI triage. Source
- **NimbusPay** - Support agent with PII redaction, guardrails, evaluations and tracing. Python, PostgreSQL. Live demo
- **Multi-Agent Travel Planner** - LangGraph planner with parallel subagents, critic loop and human approval. Source
- **DocExplainer and ExpenseTracker** - RAG and tool-calling agents on .NET 8, Next.js and pgvector. Source
- **RideSmart (Agent, Android, Mobile)** - Ride price comparison: Python / Playwright, native Kotlin / Compose reading live fares, Expo / React Native. Source

All projects, with architecture write-ups: varshini0511.github.io

EDUCATION AND LANGUAGES

B.E. Electrical and Electronics Engineering - Mepco Schlenk Engineering College (Autonomous), Sivakasi
Languages: English, Tamil, Telugu